Depu Meng, Ph. D.

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Research Engineer at Applied Intuition

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Research Interests

World foundation model: Robotics world model, video generation, world action model, multi-modal foundation model pretraining, end-to-end autonomous driving.

Education

University of Science and Technology of China - Microsoft Research Asia — Beijing, China

Ph. D. in Control Science and Engineering — *Sept. '18 – Jun. '23*

- Advisors: Dr. Baining Guo (Microsoft), Prof. Houqiang Li (USTC)

University of Science and Technology of China — Hefei, China

B.E. in Electrical Engineering (School of Gifted Young) — *Sept. '14 – Jun. '18*

Work Experience

Applied Intuition — Mountain View, CA

Research Engineer, Research — *Sep. '25 – present*

- Physical world model, video generation, 3D foundation model, scene reconstruction, and end-to-end driving foundation models.

DiDi Autonomous Driving — San Jose, CA

Machine Learning Engineer, AI Research — *Dec. '24 – Aug. '25*

- End-to-end driving model training and 3d reconstruction with Gaussian Splatting.

University of Michigan — Ann Arbor, MI

Research Fellow, Michigan Traffic Lab — *Aug. '23 – Dec. '24*

- Near-miss detection and road-side perception.
- Led the team won USDOT Intersection Safety Challenge Stage 1A and 1B.

Internship Experience

University of Michigan — Ann Arbor, MI

Research Assistant, Michigan Traffic Lab — *Apr. '22 – Aug. '23*

- Image generation and driving scenario simulation.

Mentor: Prof. Henry X. Liu

Meituan — Beijing, China

Intern, Autonomous Delivery Group — *Aug. '21 – Apr. '22*

- Joint Lidar-based object detection and motion prediction.

Mentor: Dr. Changqian Yu

Microsoft Research Asia — Beijing, China

Intern, Visual Computing Group — *Jul. '19 – Jul. '21*

- Transformer-based object detection and unsupervised pretraining.

Mentor: Dr. Jingdong Wang

Microsoft Research Asia — Beijing, China

Intern, Visual Computing Group — *Jul. '17 – Jul. '18*

- CNN architecture design and image classification.

Mentor: Dr. Jingdong Wang

Publications

⋆ equal contribution, † project lead.

LaMo: Self-Supervised Latent Motion Priors for Physical Realism in Video Generation
Bo Jiang, **Depu Meng**, Yihan Hu, Yichen Xie, Tianshuo Xu, Wei Zhan
Submitted to NeurIPS 2026

Teaching Video Generators to Remember: Eliciting Dynamic Memory for Out-of-Sight State Evolution
Tianshuo Xu, Yichen Xie, **Depu Meng**†, Chensheng Peng, Quentin Herau, Bo Jiang, Yihan Hu, Wei Zhan
Submitted to NeurIPS 2026

URoPE: Universal Relative Position Embedding across Geometric Spaces
Yichen Xie, **Depu Meng**, Chensheng Peng, Yihan Hu, Quentin Herau, Masayoshi Tomizuka, Wei Zhan
Submitted to ECCV 2026

SpectralSplat: Appearance-Disentangled Feed-Forward Gaussian Splatting for Driving Scenes
Quentin Herau⋆, Tianshuo Xu⋆, **Depu Meng**†, Zhijie Yang, Chensheng Peng, Spencer Sherk, Yihan Hu, Wei Zhan
Submitted to ECCV 2026

Exploring Communication and Roadside Perception Requirements for Cooperative Warning Systems at Intersections
Tinghan Wang, **Depu Meng**, Boqi Li, Rusheng Zhang, Yukun Zuo, Shengyin Shen, Darian Hogue, Michael Maile, Michael Shulman, Henry X. Liu
IEEE Intelligent Vehicles Symposium, 2025

Towards Comprehensive Roadside Intelligence: Sensor Fusion and Full-stack Perception with Multiple Cameras
Rusheng Zhang, **Depu Meng**, Shengyin Shen, Boqi Li, Tinghan Wang, Henry X. Liu
IEEE Intelligent Vehicles Symposium, 2025

MSight: An Edge-cloud Infrastructure-based Perception System for Connected Automated Vehicles
Rusheng Zhang⋆, **Depu Meng**⋆, Shengyin Shen, Zhengxia Zou, Houqiang Li, Henry X. Liu
Tech Report.

Robust Roadside Perception for Autonomous Driving: An Annotation-free Straegy with Synthesized Data
Rusheng Zhang, **Depu Meng**, Lance Bassett, Shengyin Shen, Zhengxia Zou, Henry X. Liu
IEEE Transactions on Intelligent Vehicles, 2024

Systematic Assessment of Roadside Perception Systems for Automated Vehicles: Insights from Field Testing
Rusheng Zhang, **Depu Meng**, Tinghan Wang, Tai Karir, Shengyin Shen, M. Maile, M. Shulman, Henry X. Liu
Transportation Research Board Annual Meeting, 2024

ROCO: A Roundabout Traffic Conflict Dataset
Depu Meng, Owen Sayer, Rusheng Zhang, Shengyin Shen, Houqiang Li, Henry X. Liu
Transportation Research Board Annual Meeting, 2023

HyMo: Hybrid Motion Representation Learning for Prediction from Raw Sensor Data
Depu Meng, Changqian Yu, Deheng Qian, Houqiang Li, Dongchun Ren
IEEE Transaction on Multimedia, 2023

CORE: Consistent Representation Learning for Face Forgery Detection
Yunsheng Ni, **Depu Meng**, Changqian Yu, Chengbin Quan, Dongchun Ren, Youjian Zhao
CVPR 2022 Workshop on Media Forensics

Conditional DETR for Fast Training Convergence
Depu Meng*, Xiaokang Chen*, Zejia Fan, Yuhui Yuan, Gang Zeng, Houqiang Li, Lei Sun, Jingdong Wang
International Conference on Computer Vision, 2021

Consistent Instance Classification for Unsupervised Representation Learning
Depu Meng, Zigang Geng, Zhirong Wu, Bin Xiao, Houqiang Li, Jingdong Wang
ICCV 2021 Workshop on Self-supervised Learning for Next-Generation Industry-level Autonomous Driving

Bottom-Up Human Pose Estimation by Ranking Heatmap-Guided Adaptive Keypoint Estimates
Ke Sun, Zigang Geng, **Depu Meng**, Bin Xiao, Dong Liu, Zhaoxiang Zhang, Jingdong Wang
Tech Report

Deep Convolutional Neural Networks with Merge-and-Run Mappings
Liming Zhao, Mingjie Li, **Depu Meng**, Xi Li, Zhuowen Tu, Zhaoxiang Zhang, Yueting Zhuang, J. Wang
International Joint Conference on Artificial Intelligence, 2018

Awards

U.S. DOT Intersection Safety Challenge Stage 1B Tier 1 Winner Team (Prize: $750,000) *Jan. '25*

U.S. DOT Intersection Safety Challenge Stage 1A Winner Team (Prize: $100,000) *Jan. '24*

Shenzhen Stock Exchange Scholarship, USTC *Dec. '22*

Star of Tomorrow Internship Award, Microsoft Research Asia *Jul. '18*

First Prize in Intelligent Robot Competition, Harbin Institute of Technology *Jul. '16*

The AEGON-INDUSTRIAL Fund Scholarship, USTC *Oct. '15*

Services

Conference Reviewer: CVPR, ECCV, ICCV, NeurIPS, ACM MM, TRBAM, IROS 2023

Journal Reviewer: IJCV, IEEE T-IV, IEEE T-MM, IEEE T-CSVT, Neurocomputing, Pattern Recognition